Out-applicatives are concealed comparatives*

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Abstract

The prefix *out-* attaches to nouns, adjectives, and verbs to form transitive verbs. We argue that the resulting verbs describe an event with the subject argument as a participant and have a uniformly comparative meaning, namely that the degree of the event along some contextually fixed dimension of comparison exceeds a standard identified by the object argument. The object can identify the standard in one of three ways: (a) directly by describing it, (b) by establishing an event of the same type as the verbal event but with the object as a participant, or (c) by establishing a modal domain of worlds in which there is an event of the same type as the verbal event with the subject argument as a participant. We show that these varieties of comparison have equivalent counterparts in overt *more/-er than* comparatives and in *too/not enough* comparatives.

Our analysis depends on deriving *out*-verbs by prefixing *out-* to a base with its complete argument structure and Theta-grid. We present evidence that supports this morphological derivation over Ahn's (2022) syntactic derivation, which gets the argument structure of *out*-verbs entirely from the prefix *out-* independently of the base's argument structure. Our analysis explains in particular why the base of *out-* cannot be an obligatorily transitive verb. We defend our unified analysis against previous proposals, in particular that *out*-verbs introduce a causative or "competition" macro-event and lie on an "interpretational cline" between comparative and resultative meanings (Kotowski 2021, 2023) and that some of them constitute "a separate (though perhaps related) phenomenon … with a different (perhaps resultative) meaning" (Ahn 2012). We argue that the causative, resultative, and competition interpretations are inferential.

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1 The prefix *out-*

The prefix *out-* turns intransitive verbs, adjectives, and nouns into transitive verbs.[1]

(1) a. A baby caribou can outrun its mother by the age of three days.
b. Andrew Cuomo was outsmarted by bartender Alexandria Ocasio-Cortez.
c. CBS certainly outgeneraled and outadmiraled its rivals.

Out-prefixation is productive. Inventive nonce words like (2) (from the Internet unless otherwise marked) are readily understandable:

(2) (a) some Dutch ladies, out-gutturaling even the Swiss themselves (OED)
(b) A Parsee team to outcricket an English eleven! (OED)
(c) We must out-Game-of-Thrones him by making powerful alliances.
(d) It was all this big elaborate joke to her, out-Southerning the folks who made a big deal about such stuff.
(e) People tend to surround themselves with people who don't outskill them when they outteam them. (OED)
(f) I had the glory of out-long-wording both parties. (OED)
(g) On the inside, Okinawa sweet potatoes can outpurple an eggplant. (OED)
(h) It is no longer Navratilova's world when she can not only be outplayed but also outsyllabled. (OED)
(i) He was sure their clients out-healed, out-helping-handed, overall out-charitied their competing charities.

These verbs express a comparison on some quantitative or qualitative scale between two events[2] – one in which the underlying subject argument is a participant and a similar one in which the underlying object argument is a participant (passivized in (2h)). The scale can be inferred from the base, possibly with contextual information (e.g. *outcompete* = *outdo in competition*). When the base is a noun or adjective, as in (2), the event

[1] Unless otherwise specified, all examples are from the Internet, sometimes abbreviated for readability.

[2] We use *event* in the broader sense of 'eventuality', which includes states and processes. The events could of course be part of a larger event, as in *Fischer outplayed Spassky in their 1972 match.*

itself must be inferred. Language users converge intuitively on the interpretation of creative uses of *out-*. Without additional context, *out-long-wording* is understood to mean "uttering more long words" – not learning more of them, or avoiding more of them, for example. *Out-syllabled* in the context of (2h) means "containing more syllables", not "uttering more syllables". *Out-Game-of-Thronesing* means acting more deviously in the manner of *Game of Thrones*, not watching more episodes of it on TV.

Such verbs are numerous and freely coined, yet morphologically and syntactically highly constrained. *Out-* cannot be prefixed to adverbs, conjunctions, or prepositions, or to obligatorily transitive verbs, or to verbs that require an obligatory predicate complement:

(3) **outsoon, *outquickly, *outbecause, *outunder, *outnotwithstanding*
*outresemble, *outinvolve, *outhave, *outconsider, *outentail, *outimply, *outdespise, *outdetach, *outrepair, *outenjoy, *outreplace, *outrecommend, *outdeploy

*outbe, *outseem, *outshould, *outcan

For example, **John outresembled Fred* and **Biden outwas Trump confused* are ungrammatical, or even uninterpretable.

We will argue that *out-*verbs are morphologically derived by attaching the prefix *out-* to the base to form a transitive verb. The resulting verbs preserve the arguments and Theta-roles of the base, if it has any, and have a comparative meaning. The prefix and base are combined in the following way. If the base has neither a subject nor an object argument, as in the case of nouns and adjectives, two arguments are added to form a transitive verb, e.g. *outmuscle, outfox, outdistance, outsmart, outmodern*. If the base is an intransitive verb, its subject argument is retained, and an object is added, e.g. *outjump, outsing, outwrestle*. The base then identifies the compared events, and the arguments identify their respective subject roles. If the base is a transitive verb, its object argument must be omitted. So *outwrite, outdrink, outeat, outresearch, outearn, outcompete, outreproduce, outspend, outlobby, outgoogle, outcook, outhit, outpunch, outgive, outperform, outinfluence, outdonate* are derived from the objectless versions of the base verbs. If a transitive base cannot be interpreted without its object, and the object cannot be ellipsed without a parallel context (what we will call a *rigid* object argument), it cannot be prefixed with *out-* at all (Bresnan 1982). For example, *John outwrites Bill* is grammatical for the same reason that *John writes (well)* and *John is a writer* are grammatical, and **John outresembles Bill* is ungrammatical for the same reason that **John resembles* and **John is a resembler* are ungrammatical. For analogous reasons *out-* cannot be prefixed to verbs with rigid predicate complements, such as *be, seem, appear*: *Biden was more confused than Trump* cannot be expressed as **Biden outwas Trump confused*.

The prohibition that bars prefixation of *out-* to transitive verbs is not arbitrary. It is the consequence of a grammatical/semantic bottleneck. *Out-* requires two arguments, one for the comparandum (the event that is compared), the other for the comparand (the event that it is compared to). But a rigidly transitive base has an additional obligatory argument. So *out-*comparatives from rigid transitives would have three obligatory direct arguments. If one is left out, the sentence is uninterpretable: **John outadmires Mary* makes no sense because *Mary* can't be *both* the admirer who is outdone by John *and* the admiree. But the syntax has no room for all three. **John outadmires Mary than Fred* is syntactically ill-formed. So is **John outadmires Mary Fred*, for the first object of a ditransitive must be a recipient-type argument. Alternatives like *John outadmires Mary with/for Fred* are excluded because the prepositions cannot convey the intended semantic relation.

The pattern (4) reflects the same bottleneck:

(4) a. (i) We drink wine. (ii) *We outdrink wine.
b. (i) We drink. (ii) We outdrink you. (iii) *We outdrink you wine / wine you.

Contrasts like (5), between *appreciate* (i) 'to value', and (ii) 'to increase in value' are also predicted.

(5) a. (i) We appreciate you/*$\varnothing$. (ii) *We outappreciate you/$\varnothing$.
b. (i) Real estate has appreciated. (ii) Real estate has outappreciated commodities.

There is however one class of rigidly transitive verbs to which *out-* can be prefixed. These are verbs that require measure phrases as complements, such as *weigh, yield, gross, spend*. To the extent that measure phrases are DP-like, in that they can have articles (at least indefinite ones) and adjective modifiers, we can consider them objects and classify such verbs as transitive.

(6) a. (i) Fred weighs a good 200 pounds. (ii) He outweighs Max by an estimated 80 pounds.
b. (i) An acre of wheat yields an average of 40 bushels. (ii) Fallow fields outyield nonfallow fields.

But measure phrase complements are peculiar objects in many respects. They are non-referential, cannot have definite articles (**Fred weighs the good 2000 pounds*), resist passivization (**Two years were lasted by the war, *Three days were stayed by the guests*), and cannot serve as the obligatory object argument of *out*-verbs.

(7) a. *The famine outlasted two whole years. (OK: The famine outlasted the war.)
   b. *The guests outstayed two months. (OK: They outstayed their welcome.)
   c. *Harris outraised $1 billion. (OK: Harris outraised Trump.)

With comparatives, measure phrases provide a differential expressed by prepositional phrases. The same is seen with *out*-, as illustrated in (6). The reason why *out*- can be prefixed to verbs like *weigh* is that their measure phrase objects can be rephrased as prepositional measure phrases, leaving the object position free for the comparand's subject, as in the (ii) examples in (6). But, as illustrated in (7), measure phrases cannot be objects of *out*-applicatives, presumably because of their syntactically defective nature.[3]

The highly regimented morphology and syntactic uniformity of *out*-verbs contrasts with their semantic flexibility. Many *out*-verbs are polysemous. Their interpretation depends on how the dimension of comparison is specified and on how the direct object determines the threshold for the comparison. Consider the sentence *Mary outgrew her stoner boyfriend*. It can be an event comparison (8a), a degree comparison (8b), and – in this case the most accessible reading – a modal degree comparison (8c).

(8) Mary outgrew her stoner boyfriend.

   a. *Event comparison*
      i. 'Mary grew faster than her stoner boyfriend did.'
      ii. 'Mary grew by a larger amount than her stoner boyfriend did.'
   b. *Degree comparison*

      'Mary grew bigger than her stoner boyfriend.'
      [Can be true even if her boyfriend did not grow at all, or even shrank.]

[3]As discussed in section 3, degree descriptions, by contrast, are perfectly good objects for *out*-applicatives.

c. *Modal degree comparison*

'Mary grew too old (or too mature) for her stoner boyfriend.' = 'Mary reached an age (or level of maturity) greater than one in which she could have (or wanted to have) the stoner as her boyfriend.'

We will refer to the first and second terms of a comparison as the *comparandum* and the *comparand*, and for want of a better term, to the added direct object argument introduced by *out-* as the *applicative* argument. The applicative argument determines the type of the comparand and specifies a degree which sets a threshold that is exceeded by the comparandum.

The addition of the applicative argument, the prohibition of transitive bases, and the special behavior of measure phrases seen in (6) and (7), all confirm that *out*-applicatives are covert comparatives. *Out*-applicatives are of several types. In the following sections we lay out how these comparative meanings come about.

2 Event comparatives

In the *event comparison* reading exemplified by (8a), the subject and object arguments are participants in two events of the same type – in this case, growing events. The *out*-verb then expresses a comparison between the maximum degree of two events along some dimension (such as time, distance, quality, or number) that is compatible with the base predicate. The event whose participant is the object argument sets a threshold for the event whose participant is the subject argument – here the amount of growth or the rate of growth. See the additional examples in (9).

(9) a. She is estimated to outearn her husband by $4 million. [= earn more money]
b. The stores in Paris outdraw both the Louvre and the Eiffel Tower. [= draw more customers]
c. She could outdraw any member of the gun club. [= draw her gun faster]
d. The sun outshines the moon by a ratio of 1 to 400,000.
e. Stocks outperformed bonds.
f. The Trump administration outdoes itself on climate change denial.
g. Hispanic women outvoted Hispanic men 52% to 48%.

A characteristic of event comparatives is that they imply the existence of two events whose degree along some dimension is compared – a comparandum and a comparand, with distinct participants. They have equivalent overt comparative paraphrases with *more/-er*.

(10)
a. Smith far outdistanced Jones. [= covered more distance]
b. Invasive Mahonia plants outnumber their native relatives. [= are more numerous]
c. Hyperdominant arboreal species outgrow others in the rainforest whenever space opens up. [= grow faster / more vigorously]
d. Mary outbid John in the auction. [= bid more]
e. The planet outlived its sun by a million years. [= lived longer]
f. Kim easily outclassed everyone in dancing. [= showed more class]
g. The sun outshines the moon by a ratio of 1 to 400,000. [= shone more strongly]
h. Do-nothing House of Representatives outdoes itself again. [= does even less = is even more inactive (than usual)]
i. Hispanic women outvoted Hispanic men. [= voted more]

It appears that each reading of an event-type *out*-comparative replicates a reading of an explicit verbal comparative. For example, (10i) has the same readings as *Hispanic women voted more than Hispanic men voted*:

(11)
a. Proportional reading: The proportion of Hispanic women that voted was greater than the proportion of Hispanic men that voted.
b. Cardinality reading: The number of Hispanic women that voted was greater than the number of Hispanic men that voted.
c. Frequency reading: Hispanic women voted more often than Hispanic men voted.

Moreover, the meanings of verbal comparatives can correspond to nominal comparatives based on their arguments. For example, *More Hispanic women voted than Hispanic men did* can mean both (11a) and (11b).

Moreover, *out*-applicatives obey the *commensurability condition*, which has been shown to hold for overt comparatives (Kennedy 1997, Doetjes 2010). The commensurability condition requires that the compared events be of the same type, and that the participants engage in the same type of activity in them is mirrored exactly by *out*-applicatives.

The parallelism between *out*-applicatives and overt comparatives goes still deeper. When compared events are not identical but sufficiently similar, an indirect interpretation can be constructed. For example, although volcanoes don't run, *John ran faster than the volcano* is a felicitous way of comparing the speed of John's running with the speed of the lava flow. It is licensed by a coercion mechanism that replaces the predicate of the comparandum with a superordinate predicate applicable to both events. Exactly the same coercion process also applies in *out*-applicatives, such as *John was able to outrun the volcano*. When the subject and object cannot be participants of an event of the same type, commensurability can be achieved, as in overt comparatives, by replacing the predicate of the comparandum with a superordinate predicate of which both are hyponyms. The parallelism between covert and overt comparatives with respect to this commensurability strategy is illustrated in (12).

(12) (a) (i) John outran the eagle = John ran faster than the eagle (moved/flew/*ran).
(ii) *John outflew the eagle = *John flew faster than the eagle (moved/flew/ran).

(b) (i) The eagle outflew John = The eagle flew faster than John (moved/ran/*flew).
(ii) *The eagle outran John = *The eagle ran faster than John (moved/flew/ran).

(c) (i) Can you outrun a volcano? = Can you run faster than a volcano (*runs/flows)?
(ii) *Can a volcano outflow you? = *Can a volcano flow faster than you?

(d) (i) The shark outswam the boat = The shark swam faster/better/longer than the boat (moved/swam).
(ii) *The boat outswam the shark = *The boat swam faster/better/longer than the shark (moved/swam).

The (i) sentences are acceptable, while their (ii) counterparts are degraded to an extent that may vary between speakers, but is invariant across covert and overt comparatives. The hallmark of coercion is that it arises when the straightforward reading is unavailable for pragmatic reasons or from background knowledge. For example, *John outplayed Mary* means "John played better/louder than Mary played" unless background knowledge tells us that Mary did not play, in which case we try to accommodate the interpretation in the best possible way, perhaps by inferring that she may have sung. When no such hypernymic relation between the compared events is found, coercion fails. For example, **John outplayed the beer* is anomalous to the extent that we can't find a V for "John played better/more than the beer V-ed" (*performed/*tasted…).

A particularly complex case of coercion is (13):

(13) I will outlive this danger, and outdie it, if I can. (C. Dickens, *Bleak House*, 1853)

Here *outlive* means 'live longer than' and *outdie* means 'die later than'. In effect, *live* and *die* are coerced into the same type of gradable predicate.

Assuming the degree-based analysis of comparatives, we can characterize the event-comparative reading of *out-* as in (14), where Q_S is a contextually recoverable gradable property of events relative to a scale S, and P is an event predicate corresponding to the verbal predicate or an appropriate hypernym.

(14) $\lambda P \lambda y \lambda x \lambda e.\, \exists d[P(e,x) \wedge Q_S(e,d) \wedge d > \mathit{Max}(\lambda d'.\, \exists e'[P(e',y) \wedge Q_S(e',d')])]$

3 Degree comparatives

Whereas event comparatives compare two events of the same type, degree comparatives, such as (8b), assert that the event denoted by the verbal predicate has a contextually salient gradable proerty to a degree of an event that exceeds a threshold which is identified by the object argument, either directly or via coercion. For example, in the degree comparative reading of (8b) *Mary outgrew her stoner boyfriend*, Mary has to grow bigger than her boyfriend even if the boyfriend did not grow at all, and even if he shrunk.

In the simplest kind of degree comparative, the object argument is just a degree-denoting expression that sets the threshold that is exceeded by the event described by the verbal predicate. The examples in (15) have a natural reading as degree comparatives, as the overt comparative paraphrases confirm.

(15)
a. Active managers seek to outperform their benchmarks. [= perform better than their benchmark = reach a level that is above the benchmark level]
b. A 5-point means the team has to outscore the handicap. [= to score more than the handicap = get a score that is higher than the benchmark score]

c. How can house prices outgrow the average income? [= become greater than the average income]

d. Graduates from elite schools outperformed the norm in small companies.

e. We have then outperformed and outachieved another person's level.

The interpretation is contextually dependent. For example, the meaning of (16) depends on whether the targeted goal of BAT level water consumption is understood as a minimum or maximum level.

(16) NLMK Group outachieved the BAT level of water consumption as early as 2010.

In the more complex types of cases represented in (17), the object does not denote a degree directly, but contextually by coercing the object into a degree via a measure function that is based on the same scale as the scale of the gradable property that the described event is inferred to have.

(17) a. I have outwalked the furthest city light. [= walked farther than the farthest city light] – Robert Frost

b. My brave little doggy is trying to outbark the thunder but he's not winning.

c. the Mac screen isn't strong enough to outshine the daylight.

d. the joyful sense of safety to the hundreds on board found vent in a shout that out-shouted the storm. (OED)

We can characterize the degree comparative reading of *out* when the direct object is a degree denoting description as in (18a), and when the direct object is individual-denoting, where μ_S is a measure function based on the same scale as the scale of the gradable property Q.

(18) 1. $\lambda P \lambda y \lambda x \lambda e. \exists d[P(e,x) \ \wedge \ Q_S(e,d) \ \wedge \ d > y]$

2. $\lambda P \lambda y \lambda x \lambda e. \exists d[P(e,x) \ \wedge \ Q_S(e,d) \ \wedge \ d > \mu_S(y)]$

4 Modal comparatives

In reading (8c), the object argument likewise serves to specify a degree that Mary's growth exceeded, but this time it is specified *modally* in terms of Mary's preferences. Although it is the most complex reading of all, it happens to be the one that *outgrowing the boyfriend* most immediately evokes, by far the most frequent use of *outgrow*, and the most creative and possibly most productive type of *out*-comparative.

This type of interpretation comes about when the applicative argument refers to something that cannot be a participant of an event of the same general type as the event of the main predicate, even by predicate coercion to a hypernym. It determines the type of the comparand event indirectly by establishing a modal domain of worlds in which there is an event of the same type as the comparandum event. Modal comparatives compare the degree along some dimension of an event in the *actual* world with the same kind of event in a domain of *alternative* worlds. Hence the subject argument is a participant of the event in both types of worlds. The applicative argument is not itself a participant; it just determines the threshold indirectly by providing a modal domain. We refer to this as the *modal comparative*.

The modal domain consists of the set of worlds compatible with the content associated with the direct object. That content must contain information that has an implication about the degrees to which the comparand event has the contextually recoverable gradable property.

We can distinguish two cases of modal comparatives. In one case, the object argument denotes entities associated with informational content, either directly (for example, *predictions* or *expectations*), or indirectly (for example, *actuarial tables* or *polls*).

(19)
a. We have outrun our goal. [= run farther than we intended to run]
b. At 100, Carter has outlasted all expectations. [= has lived longer than anyone could have expected that he would live]
c. Trump inflates his record to see himself outperforming history [= performing better than what is expected on the basis of history]
d. Europe has outperformed the requirements of the existing UN sanctions regime. [= imposed sanctions to a greater degree than the UN requires.]

In (20a), since polls are not performers, the regular event comparative reading 'performed better than the polls performed' is pragmatically odd, and *the polls* is understood as specifying an expected or predicted degree of performance. Similarly, in (20b), where the regular event comparative reading is odd because actuarial tables don't live.

(20)
a. The candidates outperformed the polls [= performed better than what is expected on the basis of the polls ≠ *performed better than the polls performed]

b. John outlived the actuarial tables. [= lived longer than the actuarial tables predicted he would live, $\neq$ *lived longer than the actuarial tables lived]

The ambiguity of (21) further illuminates the difference between event readings and modal readings:

(21) He far out-performed his critics.

.a Event comparison: He performed better/more than his critics performed. [comparison of actual performances]

.b Modal comparison: He performed better than his critics thought/predicted he would/could perform [comparison of actual and potential performances].

The modal domain can also come about through a goal-oriented construal of the modality. In that case the modal domain consists of worlds where a salient goal is achieved, and which are exactly like the actual world with respect to a set of facts. The modal domain in effect specifies the prerequisites on the degrees to which the comparandum event has the contextually determined gradable property for the goal to be realized. The applicative argument helps to determine the relevant goal, or a relevant fact that figures in what is required for the goal to be achieved.[4]

(22) a. The plant outgrew its pot. [= grew too big for its pot (to fit its pot). = grew bigger than the maximum size it could be in order to fit into its pot $\neq$ *grew bigger than its pot]

b. Smith outlived his money by decades. [= lived longer than the maximum time he could afford to live = lived longer than the maximum time he could support himself with the money he had $\neq$ *lived longer than his money lived.]

c. The global economy is outgrowing the capacity of the earth to support it. [= growing too big for the capacity of the earth to support it = bigger than what the maximum size the earth can sustain.]

d. You can't outrun a crappy diet. [= You cannot run more intensively than your crappy diet requires of you to run to stay fit. $\neq$ *run more intensively than your crappy diet runs.]

[4] Examples (22d,e,f) are from Kotowski 2023.

e. You can't outcompute the market's vagaries. [= You can't compute better than the vagaries of the market require you to compute (in order to make a profit) $\neq$ *compute better than the vagaries of the market compute.]

f. …out-witting and out-sneaking the fierce-looking rat guards. [= 'sneaking with more skill than the minimum amount of skill needed to evade the rat guards'.]

Pots don't grow, requirements don't perform, but they identify a degree which sets the threshold for the growth and performance. Modal comparatives based on informational direct objects have regular *more/-er* paraphrases, but those that are based on a goal-oriented construal of the modality can be paraphrased by *too* and *not enough* constructions, whose modal comparative readings have been established by Nadathur (2019).

5 The uniqueness and origin of *out*-comparatives

In aggregate, the *out*-applicative is the only morphological construction of its kind in English, and it is a rarity in the languages of the world (we are actually not aware of any other instances of it). But it shares almost every one of its individual properties with at least one other construction of English.

Homonymous with *out*-applicatives are locative verbs like *outsource, outflow, outhaul.* In earlier English this class included numerous compounds of verbs with the normally separable particle *out-* such as *outjeer* 'jeer out', most of which are now obsolete. Kotowski (2023) makes a case for unifying *out-* in a larger comparative/competitive/locative scalar category. In this essay we treat applicative and locative *out-* as functionally distinct morphemes. Unlike applicative *out-*, locative *out-* does not verbalize nouns, but nominalizes verbs *(outlook, output, outcome, outgrowth, outreach)* or adjectivalizes them *(outdoor, outboard)* or else preserves the base's category *(outpatient, outbound).* In any case, locative *out-* verbs have no comparative meaning.[5]

[5] A tricky case is *outflank*, as in *The cavalry outflanked the enemy*. It implies no event comparison (the enemy could be stationary, or even asleep), nor a degree comparison (*the enemy* cannot be construed as a threshold-setting measure phrase), and there seems to be no modal interpretation that could yield an indirect comparison between two events. To *flank* something is to be located at its flank, and to *outflank* it is to move past its flank. Its extended use in the sense "to defeat" (as in *a determined majority of the council may outflank the direct legislation supporters*, OED) would then be a metaphorical use based on the locative meaning.

The decline of locative *out*-verbs coincides with the rise of *out*-applicatives beginning at the turn of the 17th century.

Applicative *out-* is not the only category-changing prefix that make transitive verbs by adding syntactic subject and object arguments if the base does not have them. Others include privative *de-*, repetitive *re-*, *dis- (disillusion, disrobe), en- (encourage, encircle, ensure, endanger, envision), be- (befriend, bedew, behead).* Old English and Middle English had a small number of covert comparative verbs, derived not with *out-* but with *over-*, combined with verbs (never nouns or adjectives) into inseparable prefixed verbs. The most frequent of these was *oferlyfan*, a counterpart of Latin *supervivere* 'outlive', 'survive', and probably a calque from it, like French *survivre* and German *überleben*.

(23) a. Ðet lond … wes becueden Osbearte…, gif he Cyneðryðe oferlifde.
the land was bequeathed to Osbeart…, if he Cyneðryð outlived
'The land was bequeathed to Osbeart, if he were to outlive Cyneðryð.' (early OE, Kentish dialect)

b. Seint Bryde … overlevede him by sixty ʒere. (1387)
'St. Brigid survived him by 60 years.'

c. Wilfridus … overpasseþ Beda his tyme (1387)
Wilfrid overpasses Bede his time [Latin: transcendit tempora Bedæ].
Wilfrid lives past Bede's time

d. Howe lange Marie ouerlyved hire sons ascensioune. (c1429)
'How long Marie lived past her son's ascension.'

e. ʒif deþe wolde resayve mede, Many wolde oþer ouerbede. (c1400)
'if death would receive a reward, many would overbid each other.'
(Cf. German *überbieten*)

The same prefix *over-* was also used to make verbs of excessive degree, such as *overflow, overeat, overcharge, overestimate, overfull, overgrown*, which are still in use.

(24) a. þæt land middeweard oferfleow mid fotes þicce flode. (Orosius)
'the middle of the land overflowed with a foot-deep flood.'

b. Whereas there be some men which ouerreach and goe beyond this marke. (1569)

c. God would haue a Sabbath kept; they ouer-keep it. (1608)

Along with its antonym *under-*, comparative *over-* its still in use. Unlike *out*-applicatives, they are not category-changing and they are exclusively modal and paraphrasable by *too (much)* and *not enough*. They apply a measure function to an eventuality and assert that its degree value exceeds or falls short of a threshold, which may be conventional or contextually determined or specified by a measure phrase; compare the meanings in (25):

(25) a. i. Brussels overregulates us. [= Brussels regulates us too much – one event]
ii. Brussels out-regulates London. [= Brussels and London regulate – two events]
b. i. John oversold Bill. [John touted Bill too much – one event]
ii. John outsold Bill. [John sold more than Bill sold – two events]

Over- and *under-* when used in this sense are freely added to transitive bases and do not transitivize intransitive ones: *oversleep, overreact, overprotect, overkill.*

Like all event-constructing derivatives from nominal bases, *out*-applicatives restrict their nominal base to its canonical function.

(26) If an action is named after a thing, it involves a canonical function of the thing (cf. Kiparsky 1997).

Consider "putting" verbs like *saddle* and *corral.*

(27) a. They saddled the horse.
b. They corralled the horse.

(27a) does not mean just 'They put a saddle on the horse.' It means that it is done in the way that saddles are meant to be used. For example, it cannot describe an act of loading a basket onto the horse's back for purposes of transporting it, or putting a saddle on the wrong part of the horse's anatomy. And (27b) does not mean just 'They put the horse in the corral.' It must be done in the canonical way that corrals are intended for: for one thing, the horse must be alive.

In sum: all individual features of applicative *out-*, including its affixation to nouns, adjectives, and verbs, its verbalizing and transitivizing function, and its comparative semantics, have existed since Old English times. *Only their combination is new.*

There are very few instances of *out*-applicatives before the end of the 16th century. The earliest are *outlive*[6] (replacing older *overlive*) and *outproffer* 'outbid'.[7] Palsgrave's 1530 textbook and lexicon of French cites five of them, glossed with French counterparts that seem to be otherwise unattested, at least in the relevant meaning.

(28) a. I OUTCRY. *Je forcrie*, prim. conj. Lette hym crye as loude as he wyll, yet I wyll outcrye hym: *quil crie aussi hault quil vouldra, je le veulx forcrier* or *oultrecrier*. (Palsgrave, *Lesclarcissement de la langue francoyse*, 1530)

b. I OUTEATE. *Je formangeus*, verbum defectivum, conjugate in *je mangeus*, I eate. *My horse wyll outete such four jades as thyne is*: *mon cheval formangera quatre telles charoignes quest la tienne.*

c. I out go one, I go faster than he can go. Je vas plus viste, conjugate in «I go». Though thou be goynge an hour afore me, yet I wyll out go the: *combien que tu te mets a cheminer une heure deuant moy si te oultrepasseray je*, or *si yray je plus viste que toy.*

d. I OUTRYDE. *Je oultre chevauche,* prim. conj. and *je surpasse en chevaulchant*. Take as swyfte a geldinge as thou canste fynde and I holde the twenty nobles I outryde the: *prens aussi viste hongre que tu peulx trouver, et je gaige a toy vingt angelotz que je te surpasseray en chevaulchant.*

e. I OUTRYME. *Je outre rysme,* prim. conj.

We suspect that the prefixed French verbs such as *forcrier* and *oultrecrier* that Palsgrave uses to render *out*-applicatives are calques on the English verbs. Calques and code-switching in the French spoken in England occurs from the fourteenth century on (Rothwell 2000).[8]

The full modern range of *out*-comparatives and their systematic differentiation from *over*- comparatives (types (23) versus (24)) seems to have been introduced by Shakespeare. In particular, he is the first author to coerce *out*-comparatives to commensurability[9] and the first to feature denominal and deadjectival *out-* (*out-Herod, outvenom, outvillain, outpeer, outnight, outlustre, out-tongue, outbrave, outworth, outcrafty*). In fact, it may have been this category-changing force that drove the spread of *out*-applicatives, as

[6] In cas hereafter it happen You…to outleve our…Sovereigne Lord (1472).

[7] Eyther ouzt proferyd other (1513) 'each outbid the other'.

[8] We are grateful to Eve Clark and Carol Harvey for this information.

[9] *Marke … How he outruns the wind* (*Venus & Adonis*).

the the pairing of deverbal *overstare* and deadjectival *outbrave* in (29a) suggests. Shakespeare also introduced *out-* in cognate object constructions of the type (29b) and (29c), crafted the gorgeous line (29d), and used an *out*-applicative to boast of his prowess in (29e).

(29) a. I would *ore-stare* the sternest eyes that looke:
Out-braue the hart most daring on the earth. (*Merchant of Venice*, ca. 1596)

b. Our prayers do outpray his. (*Richard II*, ca. 1595)

c. it out Herod's Herod (*Hamlet*, ca. 1600)

d. My selfe could else outfrowne false Fortunes frowne. (*King Lear*, ca. 1605)

e. Not marble, nor the guilded monument,
Of Princes shall out-liue this powrefull rime. (*Sonnet* 55)

6 Preservation of the base's argument structure

Previous treatments of *out*-verbs recognize the semantic heterogeneity of *out*-verbs, but theorize it in different ways. Most of them concentrate on event comparatives like (8a), and to the extent that they treat plain degree and modal comparatives like (8b,c) at all, they classify them as resultatives. In a tightly reasoned syntax-centered analysis of event comparative *out*-applicatives in the framework of Distributed Morphology, Ahn (2023) argues that they have a comparative meaning and that their argument structure is independent of the base's argument structure, and comes entirely from the prefix *out-*. He touches only briefly on the degree and modal event *out*-applicatives, suggesting that, contrary to what we have argued here, they are not comparatives, but have a different, perhaps resultative meaning.

Ahn's derivation of *out*-applicatives merges *out-* with a minimal VP that has no superordinate syntactic structure, erasing the argument and introducing two new arguments which do not saturate any Theta-roles of its base. Rather, the base of *out-* constrains the dimension of comparison semantically, and the types of events in which the two arguments of *out-* can be interpreted as participating in is constrained by conceptual/pragmatic reasoning in which syntactic or semantic structure plays no role (Ahn, p. 431).

As one piece of evidence for this analysis Ahn cites *out*-verbs like *outrun*, which seem to have a subject of a type that the basic verb does not allow, in which case it would have to be introduced by *out-*.

(30) (a) Aircraft carriers can outrun almost any other boat.

(b) *Aircraft carriers can run.

These cases require a closer look. The verb *run* denotes not only a specific type of gait (moving quickly on foot with both feet leaving the ground at each step), but many other kinds of continuous and typically rapid movement, including the movement of ships and other conveyances, machines, liquids, fingers over a surface, etc. The problem with (30b) is that it is nonsensical on the 'gait' reading and vacuous on the general 'movement' reading. Adding a manner or direction to modify the movement makes (30b) contentful and acceptable, as examples like (31) (from the Internet) testify:

(31) a. The best of the Soviet submarines can run as fast under water as the Navy's fastest ships can move on the surface.

b. U.S. aircraft carrier runs away when encountering CCP military exercise? (heading)

c. ...when the ship runs from deep into shallow water the kinetic energy stored in the entire hull converts itself into suction-energy.

d. If you think for a minute that a big ocean liner can run up alongside the wharf at any port she happens to come to...

e. The ship runs from the Texas-Mexico border all the way around to the Florida Keys.

Since *outrun* actually selects the same types of subjects as *run*, we conclude that it retains the subject of its base just like other *out*-verbs, exactly as our analysis predicts.

Ahn's second example of *out-* contributing a subject argument that supersedes the base verb's selectional requirements is *outrain*. The ungrammaticality of (32a) seems to show that weather verbs like *rain* have neither a subject argument nor an object argument. If so, then the subject and object arguments of *outrain* (32b) must be introduced by *out-*.

(32) a. *Atlanta rained. *It rained Atlanta.

b. Atlanta also out-rained Seattle in 1922 and 1923.

This is actually consistent with our analysis. For as we have already shown, *out-* supplies a subject to a subjectless base, as well as an object to an objectless base. That is how it makes transitive verbs from nouns and adjectives, which have neither, e.g. *outpace, outmuscle, outnumber, outsmart, out-Einstein*. So on the supposition that *rain* has neither a subject nor an object, *out-* will endow it with both.

That said, a slightly deeper dive into *rain* (and other weather verbs) yields a more interesting picture. First, sentences like (32b) *Atlanta out-rained Seattle* are not accepted by all speakers. Secondly, they are matched with (33b)-type examples, which are also acceptable to some but not all speakers.

(33) a. Scotland rains more and Ireland rains more than Scotland.
https://bulbagarden.net/threads/random-messages.102920/page-2408

b. Most of Scotland rains all the time, so an electric tumble dryer is a necessity.
https://www.vacuumland.org/cgi-bin/TD/TD-VIEWTHREAD.cgi?24213

c. Seattle rains more days then it doesn't.
https://www.jessikachristinephoto.com/blog-1/urban-couple-phot

d. Even in the summer, Ireland rains a lot.
https://travellingjezebel.com/one-day-in-dublin/

e. British Columbia's West Coast rains more than 50% of the year.
https://www.instagram.com/reel/C7fFSTxyME6/

We have found that speakers who accept sentences like (32b) also tend to accept sentences like (33), and allow *rain* with a locative subject as long as a measure phrase is present or implied, both in the simple verb (33) and in *out*-verbs like (32b). *Rain* would then not only be consistent with our thesis, but support it positively against the alternative that *out-* has no access to the base's argument structure. But, to repeat, even if *rain* were argumentless, *out-* would turn it into a transitive verb, as it turns nouns and adjectives into transitive verbs. On that supposition, no co-variation between (33) and (32b) woul be expected.

In any case, neither *outrun* nor *outrain* show that *out*-verbs do not inherit the base verb's subject argument but get their subject argument from *out-*. That leaves Ahn's claim without supporting evidence. The weaker claim that *out*-applicatives inherit only the base verb's subject argument but not its object argument would not be consistent with Ahn's way of implementing the argument-structural impoverishment of the input by restricting it to a small syntactic constituent, for there is no constituent that includes the base verb and its subject but not its object.

Counterevidence to Ahn's claim that *out-* discards the base's argument structure comes from the abovementioned generalization that it can't be added to rigidly transitive verbs like *resemble*, *detach*, and *have*. This generalization requires access to the full argument structure of the base from which *out*-applicatives are derived.

The contrast between *buy* and *sell* is another telling piece of evidence:

(34) a. (i) This book outsells the Bible. (ii) *This book outbuys the Bible.
b. (i) This book sells (well). (ii) *This book buys (well).

Sentence (34a) follows directly from (34b) as long as *out-* can see that transitive *sell* has a middle counterpart but *buy* does not, as our analysis predicts it can. (On the other hand, *buy* is not a rigid transitive, so we can have both *Do-it-yourselfers buy from us*, and therefore also *Do-it-your-selfers outbuy professionals 3-to-1*, as predicted.) In the derivation of (34ai), transitive *sell* is detransitivized to the middle *sell* and then retransitivized by *out-* – it could then be re-detransitivized again by passivization, e.g. *The Bible has never been outsold by any other book.*

If argument structure is not visible to *out*-affixation, we lose the explanation for this data, and specifically for the difference between *buy* and *sell* in (34). The reason is that every buying event is also a selling event and vice versa, and so the difference between *buy* and *sell* lies not in the semantics but in their argument structure – that is, in the way the semantic roles are associated with the syntactic arguments. Ahn (p. 458) notices this issue and proposes a somewhat problematic workaround to deal with it.

7 Inferential meaning: resultativity and competition

Out-prefixation has the same category-changing and transitivizing function as the resultative construction, e.g. *to floor an opponent, to down a drink, to up the ante, to cellar the wine*. Capitalizing on this similarity, McIntyre 2003 and Nagano 2011 claim that all *out*-verbs are resultatives. In their analysis, the prefix does not introduce a comparative, but a sub-event that the object argument participates in, which is caused by an event that the subject argument participates in, creating a complex resultative predicate. "The Object-argument is "outdone" or "beaten" as an effect of the Subject-argument's involvement in the event denoted by the verbal predicate. The resultative interpretation can moreover be understood as an actual or implicit *competition* in which the subject argument is the winner. We think that that the resultative and competition readings are context-dependent inferences (Barsalou 1982).

Resultatives are usually regarded as semantically complex predicates with two sub-events, in which the sub-event that the object argument participates in is caused by the sub-event that the subject argument participates in. For example, a resultative analysis of *Fred outdrank Stan* would be that there was an event of drinking by Fred which resulted in Stan being defeated. As Kotowski (2021) notes, this does not really capture the full meaning of the sentence, in particular that Stan was also drinking, and that Fred drank more than Stan did. Nor does it fit (8b,c), of course. Kotowski (2021, 2023) reviews and criticizes the comparative and the resultative analyses of *out*-verbs, concluding that the verbs lie on a cline between multiple related meanings, where the range of possible interpretations is context- and participant-bound rather than primarily driven by the base to which *out-* is prefixed.

According to Kotowski's (2021) analysis of *out-V*, the interpretation of the construction involves a *competition*: the subject argument surpasses the object argument on some dimension, thereby defeating it. The competition can be made explicit:

(35) a. He was sure their clients out-healed, out-helping-handed, overall out-charitied their competing charities.

b. It is routine for presidential candidates to try to "out-pro-Israel" each other.

c. trying to out-right-wing each other

d. to bolster political legitimacy by "out-Indianing" their opponents through support of local cultural institutions.

Such resultative and competition meanings as *out*-applicatives may have, we argue, are inferential. For example, (36a) (from a sports page) tells you not only that the Vikings rushed, passed, and threatened more than the Bears did, but that they competed in a game, and that the Vikings won.

(36) a. The Vikings outrushed, outpassed and outthreatened the Bears.

b. Sebastian Coe outran Roger Bannister.

c. The pronghorn outdistanced the cheetah.

(36b) implies that Coe and Bannister were vying to reach a common goal at the same time, rather than just moving independently at different times; the background knowledge that they actually never competed or raced at the same time makes it anomalous. (36c) suggests that the pronghorn was chased by the cheetah and escaped it handily, rather than just by the skin of its teeth. Even though a competition scenario is far-fetched for *outlast* and *outweigh* in (6), they still seem to convey that the quantities involved, and the differences between them, are somehow significant, in a way that their explicit comparative counterparts *last longer* and *weigh more* do not.

However, these are defeasible implicatures, for *out*-verbs can be used when there is no competition, as with non-sentient participants, and where the difference in degree between the events can be explicitly specified as negligible:

(37) a. this bat species is one of the smallest, growing only big enough to barely outweigh one US nickel.

b. The utilization of external calcium ions at the neuromuscular junction is restricted to a brief period which barely outlasts the depolarization of the nerve ending.

c. Of particular interest are the systems close to the tipping point, when dispersion interactions barely outweigh or approach the strength of the other interactions.

If the degree of the comparandum event is understood in context to also be the degree required for some salient goal to be realized, and the degree of the comparandum exceeds the degree of the comparand, which is the modal threshold, then the basic comparative reading can inferentially yield both a resultative implication and an additional "competition" implication, namely that the subject argument prevails over the applicative argument. The contrast between (38a) and (38b) illustrates how the extra implicature depends on the volitionality of the *out*-verb's object.

(38) a. John outran Bill.

(i) Plain comparative: John ran faster than Bill.

(ii) Comparative with resultative implication: John had the goal of beating Bill by running faster/farther/better than him, and he succeeded.

(iii) Comparative with resultative and competition implication: John and Bill each had the goal of beating the other by running faster/farther/better than them, and John succeeded.

b. Indiana Jones outran the boulder.

(i) Comparative: Indiana Jones ran faster than the boulder.

(ii) Comparative with resultative implication: Indiana Jones had the goal of escaping the boulder by running faster than it, and he succeeded.

(iii) Comparative with resultative and competition implication (unavailable): #Indiana Jones and the boulder each had the goal of running faster than the other, and Indiana Jones succeeded.

8 Conclusions

We have ferreted out at least some of the regularities behind the productive use of the prefix *out-* and charted this little morpheme's vast semantic range. Non-locative *out-*verbs express a comparison between two degrees along the same dimension, in which the event with the subject argument as a participant exceeds a threshold identified by the object argument. Like overt comparatives, they allow measure phrases that specify a differential. They are derived by prefixing *out-* to nouns, adjectives, or verbs, and the prefixation outputs a transitive verb. The base provides, directly or indirectly, a dimension of comparison, and its argument structure determines the availability of *out-*prefixation. If the base is a noun or adjective, a subject and object argument are supplied, and if it is an intransitive verb, an object is supplied. If the verb is rigidly transitive, in the sense that it is uninterpretable without its object, *out-* cannot be prefixed, unless the object is a measure phrase which can be turned into a differential specification rendered as a *by*-phrase.

By taking into account inferential meaning, we found that all *out-*applicatives share the common thread of comparative semantics, corresponding both to that of ordinary *more*/*-er* comparatives and modal *too*/*enough* comparatives. This is a challenge to the idea that morphology is syntax, or a spellout of syntax. The reason is that plain and modal comparatives have not (at least so far) been shown to be syntactically derived from a single base structure. An added advantage of our analysis is that it allows *out-* to be added to a base with its argument structure intact, whereas Ahn's syntactic derivation requires discarding the base's argument structure, with the untoward empirical consequences discussed in section 6.

That said, our own analysis of *out-*applicatives still falls short of a fully unified account of their systematic ambiguities. Perhaps an underspecification approach, or alternatively the technique of ambiguity packing (Crouch 2005, 2006) could pave the way to further progress.

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